Coordination and response

Mark scheme and worked solutions for 60 questions: 30 multiple choice, 30 structured.

1 · Part 1

Core · Coordination and response · [1]

Answer: C

Explanation

The central nervous system is the brain and the spinal cord only. The nerves outside the brain and spinal cord form the peripheral nervous system, so options A and D wrongly include nerves. Sense organs contain receptors and are not part of the CNS, so option B is wrong.

2 · Part 1

Core · Coordination and response · [1]

Answer: A

Explanation

Neurones carry information as electrical impulses. Hormones are chemicals carried in the blood, not along neurones. Heat energy and oxygen molecules are not the way information is carried in the nervous system.

3 · Part 1

Core · Coordination and response · [1]

Answer: D

Worked solution

Follow the impulse from where the stimulus is detected to where the response happens: receptor (in the skin) → sensory neurone → relay neurone (in the spinal cord) → motor neurone → effector (muscle that pulls the hand away). The order of the words "sensory, relay, motor" matches "detect, pass on, act".

4 · Part 1

Core · Coordination and response · [1]

Answer: B

Explanation

Option B is right: a reflex action is an automatic and rapid response, coordinated by the nervous system, in which effectors (muscles or glands) respond to a stimulus. Reflexes are not learned (option A) and do not depend on conscious decisions (option C). Reflexes always use neurones, so option D is wrong.

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5 · Part 1 Core Coordination and response [1]

Answer: B

Explanation

The effector is the part that carries out the response, here the muscles that close the eyelid. The touch receptors detect the stimulus (option A), and the sensory and relay neurones only carry impulses (options C and D); none of these produce the blink.

6 · Part 1 Core Coordination and response [1]

Answer: D

Explanation

By definition, a synapse is a junction between two neurones. A receptor cell detects stimuli, the long fibre carries impulses along one neurone, and a bundle of neurones outside the CNS is a nerve.

7 · Part 1 Supplement Coordination and response [1]

Answer: A

Explanation

The impulse arrives first and causes vesicles to release neurotransmitter (2); the neurotransmitter then diffuses across the gap (1), binds to receptor proteins on the next neurone (4), and this stimulates an impulse in the next neurone (3). Option B has binding before diffusion, which is impossible because the molecules must cross the gap first. Option C has diffusion before release. Option D has binding before release.

8 · Part 1 Supplement Coordination and response [1]

Answer: C

Worked solution

For an impulse to cross, neurotransmitter must be released on one side and must bind to receptors on the other side. Only one side has the vesicles and only the other side has the receptors, so the "message" can only be sent one way.

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9 · Part 1 Core Sense organs [1]

Answer: D

Explanation

Receptor cells in the tongue respond to chemicals (taste). The ear responds to sound, not light; the eye responds to light, not chemicals; and the skin responds to touch and temperature, not sound.

10 · Part 1 Core Sense organs [1]

Answer: A

Explanation

The cornea refracts (bends) light as it enters the eye. The iris controls how much light enters the pupil (not focusing), the lens focuses light on to the retina (not controlling the amount of light), and the retina, not the optic nerve, contains the light receptors.

11 · Part 1 Core Sense organs [1]

Answer: C

Explanation

In bright light the pupil reflex makes the pupil smaller (its diameter decreases), and the effect of this change is that less light enters the eye than would otherwise, so the retina is protected. Options A and B wrongly have the pupil getting wider, which happens in dim light. Option D is impossible because a smaller pupil lets in less light, not more.

12 · Part 1 Supplement Sense organs [1]

Answer: B

Explanation

In dim light the pupil widens: the radial muscles contract and the circular muscles relax. The two sets of muscles are antagonistic, so they do not both contract or both relax (options A and D). Option C is what happens in bright light, when the pupil gets smaller.

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13 · Part 1 Supplement Sense organs [1]

Answer: A

Worked solution

The ciliary muscles form a ring around the lens. When the ring contracts it gets smaller, so it stops pulling on the suspensory ligaments; the ligaments go slack and the elastic lens springs into a fatter, more rounded shape. A more rounded lens bends light more, which is needed for light from a near object to focus on the retina.

14 · Part 1 Supplement Sense organs [1]

Answer: D

Explanation

Rods are more sensitive to light, so they still work in dim light, but they cannot distinguish colours; cones need brighter light. Option A reverses the sensitivity of rods and cones. Option B is wrong because it is cones, not rods, that come in three types for colour vision. Option C is wrong because rods are found away from the centre of the retina, not only at the centre.

15 · Part 1 Supplement Sense organs [1]

Answer: C

Explanation

Cones are packed most densely at the centre of the retina (the fovea), and rods are more common towards the edges. Cones are not spread evenly (option A), there are very few or no rods at the centre (option B), and there are no rods or cones where the optic nerve leaves the eye, the blind spot (option D).

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16 · Part 1 Core Coordination and response [4]

Mark scheme

(a) **[2]**

- brain (1)
- spinal cord (1)

(b) **[1]**

- nerves outside the brain and spinal cord (1)

ALLOW all the nerves / neurones outside the CNS

(c) **[1]**

- coordination and regulation of body functions (1)

ALLOW coordinates / controls / regulates (activities of) the body / links stimuli with responses

Worked solution

(a) The CNS is just two structures: the brain and the spinal cord.

(b) Everything else in the nervous system is the PNS: the nerves that run between the CNS and the rest of the body.

(c) The nervous system coordinates and regulates body functions, for example detecting a stimulus and making sure the right muscle or gland responds.

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17 · Part 1 Core Coordination and response [5]

Mark scheme

(a) **[1]**

- (in the) skin (of the foot / sole) (1)

> **ALLOW** (sole of the) foot

> **REJECT** in the spinal cord / brain

(b) **[1]**

- sensory (neurone) (1)

(c) **[1]**

- relay (neurone) (1)

(d) **[2]**

- muscle (in the leg) (1)
- contracts (to lift the foot) (1)

> **ALLOW** "pulls the foot away / up" for the response mark only if contraction is implied, e.g. "muscle contracts and lifts the foot"

> **REJECT** motor neurone as the effector

Worked solution

(a) Pain and touch receptors in the skin of the sole of the foot detect the pin.

(b) The sensory neurone carries impulses from the receptor to the spinal cord.

(c) Inside the spinal cord, a relay neurone passes the impulses on to a motor neurone.

(d) The motor neurone carries impulses to the effector, a muscle in the leg. The muscle contracts, lifting the foot off the pin. Remember the effector is the thing that actually does the response, not the neurone that carries the impulse to it.

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18 · Part 1 Core Coordination and response [4]

Mark scheme

(a) **[2]**

- (hand is removed quickly so) reduces damage to tissues / prevents (serious) burns (1)
- no time is wasted on (conscious) thinking / the response does not have to be decided on by the brain (1)

ALLOW protects the body from harm, for the first mark

(b)(i) **[1]**

- salivary gland(s) (1)

REJECT mouth / tongue / taste receptors

(b)(ii) **[1]**

- secretes / releases saliva (1)

Worked solution

(a) A reflex does not wait for you to think about what to do, so the hand moves away from the flame within a fraction of a second. Less time near the flame means less damage to skin and other tissues.

(b) Here the stimulus is food (chemicals) in the mouth. The effector is a gland, the salivary gland, and its response is to secrete saliva. This shows that effectors are not always muscles.

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19 · Part 1 Core Coordination and response [6]

Mark scheme

(a) **[2]**

- 0.21 + 0.19 + 0.24 + 0.20 + 0.21 = 1.05 (1)
- 1.05 ÷ 5 = 0.21 s (1)
- correct answer with no working scores full marks

REJECT answer without unit for the second mark

(b) **[1]**

- 0.21 ÷ 0.05 = 4.2 (times) (1)

ECF their (a) ÷ 0.05 correctly evaluated

(c) **[2]**

- knee jerk is a reflex / automatic / does not need conscious thought or a decision by the brain (1)
- shorter pathway / fewer neurones (involved) / impulses only pass through the spinal cord (1)

ALLOW ruler catching needs the eyes to detect the ruler and the brain to decide and send impulses, for one mark

ALLOW fewer synapses, for the second mark

(d) **[1]**

- muscle (1)

ALLOW thigh muscle / muscle in the front of the upper leg / quadriceps

REJECT gland / motor neurone

Worked solution

(a) Add the five times: 0.21 + 0.19 + 0.24 + 0.20 + 0.21 = 1.05 s. Divide by the number of trials: 1.05 ÷ 5 = 0.21 s. Check: the values are spread around 0.21 s, so the mean is sensible.

(b) 0.21 ÷ 0.05 = 4.2, so catching the ruler takes 4.2 times as long. Check: 0.05 × 4.2 = 0.21.

(c) The knee jerk is a reflex. Impulses go from the receptor to the spinal cord and straight back to the leg muscle, with no conscious decision. To catch the ruler, the eyes must detect it falling, impulses go to the brain, the brain decides to act and sends impulses down to the hand muscles, so the pathway is longer and slower.

(d) The effector is a muscle in the thigh, which contracts to kick the lower leg forward.

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20 · Part 1 Core Coordination and response [4]

Mark scheme

(a) **[1]**

- (electrical) impulses (1)

(b) **[1]**

- a junction between two neurones (1)

REJECT a junction between a neurone and a muscle

(c) **[1]**

- 2 (1)

(d) **[1]**

- spinal cord (1)

REJECT brain

Worked solution

(a) Information travels along neurones as electrical impulses.

(b) A synapse is a junction between two neurones.

(c) Count the places where one neurone meets another: sensory neurone to relay neurone is one synapse, and relay neurone to motor neurone is a second. The receptor and the muscle are not neurones, so the pathway has 2 synapses.

(d) In this reflex the relay neurone is in the spinal cord, which is part of the CNS. The impulse does not need to go to the brain for the hand to move.

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21 · Part 1 Supplement Coordination and response [4]

Mark scheme

(a) **[1]**

- synaptic gap (1)

ALLOW synaptic cleft

(b) **[1]**

- neurotransmitter (molecules) (1)

REJECT hormones / enzymes

(c) **[1]**

- the first neurone / the neurone before the gap / the neurone the impulse arrives along (1)

(d) **[1]**

- on (the membrane of) the next neurone / the neurone after the gap (1)

Worked solution

(a) The neurones at a synapse are separated by a tiny space called the synaptic gap.

(b) The vesicles contain neurotransmitter molecules.

(c) The vesicles are in the end of the neurone that the impulse arrives along, before the gap.

(d) The receptor proteins are on the membrane of the next neurone, on the other side of the gap. This arrangement (vesicles on one side, receptors on the other) is why impulses only cross a synapse in one direction.

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22 · Part 1 Supplement Coordination and response [5]

Mark scheme

(a) **[4]**

- impulse (arriving at the end of the sensory neurone) stimulates the release of neurotransmitter (1)
- (neurotransmitter is released) from vesicles into the synaptic gap (1)
- neurotransmitter diffuses across the gap (1)
- (neurotransmitter) binds with receptor proteins on the relay neurone (1)

ALLOW "an impulse is then stimulated in the relay neurone" as an alternative to any one of the first two marks if the sequence is otherwise correct

REJECT impulse / electricity jumps across the gap

(b) **[1]**

- takes longer, because the neurotransmitter has further to diffuse (1)
- both the direction of change and the reason are needed for the mark

Worked solution

(a) The impulse cannot jump the gap, so a chemical carries the message across. When the impulse reaches the end of the sensory neurone, vesicles release neurotransmitter molecules into the synaptic gap. The molecules diffuse across the gap and bind with receptor proteins on the membrane of the relay neurone. This binding stimulates a new impulse in the relay neurone.

(b) Diffusion is slower over longer distances, so a wider gap would mean the neurotransmitter takes longer to reach the receptor proteins, and the impulse would take longer to cross.

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23 · Part 1 Supplement Coordination and response [6]

Mark scheme

(a) **[2]**

- (drug P) has a shape that fits / is complementary to (the shape of) the receptor protein (1)
- (because) it has the same / similar shape as the neurotransmitter, which normally binds to the receptor (1)

(b) **[2]**

- neurotransmitter cannot bind to the receptor proteins / receptor proteins are blocked (1)
- so fewer / no impulses are stimulated in the next neurone / transmission is reduced or stopped (1)

(c) **[2]**

- no neurotransmitter is released into the synaptic gap / no neurotransmitter diffuses across the gap (1)
- so no neurotransmitter binds with the receptor proteins (on the next neurone) (1)

> **ALLOW** impulse cannot cross the synapse, for the second mark only if linked to lack of neurotransmitter

Worked solution

(a) Receptor proteins only bind molecules of a matching shape. Drug P has almost the same shape as the neurotransmitter, so it fits the receptor protein just as the neurotransmitter does.

(b) While drug P is attached, the receptor proteins are occupied, so the real neurotransmitter has nowhere to bind. Drug P does not itself start an impulse, so fewer (or no) impulses are produced in the next neurone.

(c) The next neurone only produces an impulse when neurotransmitter binds to its receptor proteins. If drug Q stops release from the vesicles, no neurotransmitter crosses the gap, nothing binds to the receptors and no impulse is stimulated.

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24 · Part 1 Supplement Coordination and response [5]

Mark scheme

(a) **[2]**

- impulse produced in Z (1)
- no impulse produced in X (1)

(b) **[3]**

- vesicles (of neurotransmitter) are only in the neurone before the gap / at the Y–Z synapse, Y has vesicles so it releases neurotransmitter (1)
- receptor proteins are only on the neurone after the gap / Z has receptor proteins, so neurotransmitter binds and an impulse starts in Z (1)
- at the X–Y synapse, Y has no vesicles to release neurotransmitter / X has no receptor proteins, so no impulse is produced in X (1)

ALLOW synapses ensure impulses travel in one direction only, for one mark if no other mark is scored

Worked solution

(a) The impulse reaches both ends of neurone Y, but it can only cross the synapse that leads to Z. So an impulse appears in Z but not in X.

(b) At a synapse, only the neurone before the gap stores neurotransmitter in vesicles, and only the neurone after the gap has receptor proteins. At the Y–Z synapse, Y is the neurone before the gap: it releases neurotransmitter, which diffuses across and binds to receptors on Z, so Z produces an impulse. At the X–Y synapse, Y is the neurone after the gap: it has receptor proteins but no vesicles, so it cannot release neurotransmitter towards X, and X has no receptors to respond. This is why synapses make impulses travel in one direction only.

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25 · Part 1 Core Sense organs [4]

Mark scheme

(a) **[2]**

- groups of receptor cells (1)
- (that) respond to specific stimuli (1)

ALLOW named examples of stimuli (light, sound, touch, temperature, chemicals) for the second mark

(b) **[2]**

- touch (1)
- temperature (1)

ALLOW pressure for touch; heat / cold for temperature (but not both heat and cold for two marks)

IGNORE pain

Worked solution

(a) A sense organ is a group of receptor cells that respond to specific stimuli. For example, the receptor cells in the eye respond to light and those in the ear respond to sound.

(b) The skin has receptor cells that respond to touch and others that respond to temperature, which is why you can feel both contact and heat or cold through your skin.

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26 · Part 1 Core Sense organs [5]

Mark scheme

(a) **[1]**

- cornea (1)

(b) **[1]**

- iris (1)

> **REJECT** pupil

(c) **[1]**

- lens (1)

(d) **[2]**

- optic nerve carries impulses (from the retina) to the brain (1)
- so impulses from the (light) receptors in the retina cannot reach the brain (1)

> **ALLOW** brain receives no information from that eye, for the second mark

Worked solution

(a) The cornea is the clear front of the eye; it refracts light as the light enters.

(b) The iris is the coloured ring that controls the size of the pupil and so how much light gets in. The pupil is only a hole, so it cannot control anything itself.

(c) The lens focuses the light on to the retina.

(d) The retina's receptors still turn light into impulses, but those impulses travel to the brain along the optic nerve. With the optic nerve damaged, the impulses never reach the brain, so the patient does not see.

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27 · Part 1 Core Sense organs [5]

Mark scheme

(a) **[2]**

- as light intensity increases, pupil diameter decreases (1)
- the decrease becomes smaller / levels off at higher light intensities (1)

ALLOW quoted data for the second mark, e.g. diameter falls by 1.5 mm from 10 to 100 lux but only 0.5 mm from 1000 to 2000 lux

REJECT "the pupil diameter is proportional to light intensity"

(b) **[2]**

- decrease = 7.0 − 3.0 = 4.0 mm and 4.0 ÷ 7.0 × 100 (1)
- 57 (%) (1)
- correct answer with no working scores full marks

ALLOW 57.1 (%) for the second mark

ECF an incorrect decrease divided by 7.0 and multiplied by 100, correctly evaluated

(c) **[1]**

- less light enters the eye, so the retina / light receptors are protected from damage (1)

Worked solution

(a) Read down the table: as light intensity goes up, pupil diameter goes down. The pupil shrinks a lot at low intensities (7.0 to 5.5 mm between 10 and 100 lux) but only a little at high intensities (3.0 to 2.5 mm between 1000 and 2000 lux), so the change levels off.

(b) Decrease = 7.0 − 3.0 = 4.0 mm. Percentage decrease = 4.0 ÷ 7.0 × 100 = 57.1%, which is 57% to the nearest whole number. Check by working backwards: 7.0 × 0.57 ≈ 4.0, and 7.0 − 4.0 = 3.0 mm.

(c) A smaller pupil lets less light reach the retina, which protects the light receptors from damage by very bright light.

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28 · Part 1 Supplement Sense organs [5]

Mark scheme

(a) **[3]**

- radial muscles contract (1)
- circular muscles relax (1)
- pupil gets wider / dilates, so more light enters the eye (1)

REJECT muscles in the pupil

(b) **[2]**

- (the two sets of) muscles work in opposition / when one contracts the other relaxes (1)
- (so they) have opposite effects on the size of the pupil / one makes the pupil smaller and the other makes it larger (1)

Worked solution

(a) When the room becomes dim, the pupil reflex widens the pupil. The radial muscles, which run outwards from the pupil like spokes of a wheel, contract and pull the pupil open. At the same time the circular muscles, which form a ring around the pupil, relax. The wider pupil lets more light reach the retina so Ravi can still see.

(b) Antagonistic muscles work as a pair with opposite effects. When the circular muscles contract (and radial muscles relax) the pupil gets smaller; when the radial muscles contract (and circular muscles relax) the pupil gets larger.

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29 · Part 1 Supplement Sense organs [6]

Mark scheme

(a) **[1]**

- lens (1)

(b) **[4]**

- ciliary muscles relax (1)
- suspensory ligaments become tight / tension in suspensory ligaments increases (1)
- lens is pulled thinner / less rounded / less convex (1)
- light is refracted less, so it is focused on the retina (1)

REJECT ciliary muscles contract

(c) **[1]**

- lens cannot become rounded enough, so light from near objects is not refracted enough to be focused on the retina (1)

ALLOW light from near objects is focused behind the retina / image on the retina is blurred

Worked solution

(a) Accommodation works by changing the shape of the lens.

(b) While Leila was reading, her ciliary muscles were contracted and her lens was rounded. To see the distant mountain, the ciliary muscles relax. This pulls the suspensory ligaments tight, and the tight ligaments pull the lens into a thinner shape. Light from a distant object needs less bending, so the thinner lens refracts it less and focuses it on the retina.

(c) Close-up objects need a strongly rounded lens to bend the light enough. If the lens is too stiff to become rounded, light from small print is not refracted enough to focus on the retina, so the print looks blurred.

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30 · Part 1 Supplement Sense organs [6]

Mark scheme

(a) **[2]**

- most cones / highest concentration of cones at the centre (of the retina) / at R (1)
- (very) few cones away from the centre / at P, Q and T (1)

> **ALLOW** quoted data, e.g. 150 thousand per mm^2 at R but only 5–8 thousand per mm^2 at P, Q and T, for either mark

(b) **[1]**

- (it is where) the optic nerve leaves the eye / it is the blind spot (1)

(c) **[2]**

- looking to one side makes the light fall on a region with (many) rods, e.g. Q / P / T (1)
- rods are more sensitive to (dim) light than cones / there are no rods at R, and cones are not sensitive enough for dim light (1)

(d) **[1]**

- there are three kinds of cone, each absorbing light of a different colour (1)

Worked solution

(a) Cones are packed most densely at the centre of the retina (150 thousand per mm^2 at R). Away from the centre there are very few cones, only 5 to 8 thousand per mm^2 at P, Q and T.

(b) Position S has no light receptors at all. This is where the optic nerve leaves the eye, called the blind spot.

(c) Position R has only cones and no rods. Cones need fairly bright light, so the dim star is hard to see when its light falls on R. Looking slightly to one side moves the star's light on to an area such as Q, which has a large number of rods. Rods are much more sensitive to light, so they can detect the dim star.

(d) There are three kinds of cone, and each absorbs light of a different colour. The brain compares the signals from the three kinds to produce colour vision.

1 · Part 2 Core Hormones [1]

Answer: B

Explanation

The syllabus describes a hormone as a chemical substance, produced by a gland and carried by the blood, which alters the activity of one or more specific target organs. Hormones are not enzymes and do not travel along neurones, electrical impulses belong to nervous control, and hormones affect only their specific target organs, not every organ equally.

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2 · Part 2 Core Hormones [1]

Answer: D

Worked solution

Think about what helps the man escape: faster breathing brings in more oxygen, a faster heart rate carries oxygen and glucose to the muscles more quickly, and wider pupils let more light in so he can see better. All three increase.

3 · Part 2 Core Hormones [1]

Answer: A

Explanation

Hormones travel in the blood, so hormonal control acts more slowly than nervous control, which uses electrical impulses along neurones. Hormonal effects last longer because the hormone stays in the blood for some time. The other rows get the speed, the duration, or both, the wrong way round.

4 · Part 2 Supplement Hormones [1]

Answer: C

Explanation

The pancreas secretes both insulin and glucagon. Adrenaline is secreted by the adrenal glands and oestrogen by the ovaries, so every other option includes a hormone that the pancreas does not secrete.

5 · Part 2 Supplement Hormones [1]

Answer: A

Explanation

Adrenaline increases the blood glucose concentration and increases the heart rate, so glucose and oxygen reach the muscle cells faster for respiration, which releases energy. Converting glucose to glycogen lowers blood glucose (the effect of insulin, not adrenaline), carbon dioxide is a product of respiration rather than something the muscles need, and adrenaline increases the heart rate rather than decreasing it.

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6 · Part 2 Core Homeostasis [1]

Answer: B

Explanation

Homeostasis is the maintenance of a constant internal environment. Coordinating responses to external stimuli is the role of the nervous system, removing metabolic waste is excretion, and homeostasis is about conditions inside the body, not the surroundings.

7 · Part 2 Core Homeostasis [1]

Answer: D

Explanation

Insulin decreases the blood glucose concentration, bringing it back to normal after a meal. Insulin does not increase blood glucose, adrenaline is the 'fight or flight' hormone and is not the hormone that lowers blood glucose after a meal, and testosterone is not involved in controlling blood glucose.

8 · Part 2 Supplement Homeostasis [1]

Answer: C

Explanation

In negative feedback, a change away from the set point causes a response that reverses the change, bringing the condition back to the set point. Moving the condition further away is the opposite of negative feedback, the set point itself stays the same, and no correction is needed when the condition is already at the set point.

9 · Part 2 Supplement Homeostasis [1]

Answer: A

Worked solution

Link each hormone to its job: insulin lowers blood glucose (liver stores glucose as glycogen); glucagon raises blood glucose (liver breaks glycogen down to glucose). After several hours without food the problem is low glucose, so the answer must be glucagon with glycogen being broken down.

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10 · Part 2 Supplement Homeostasis [1]

Answer: D

Explanation

Rapid, repeated muscle contractions are shivering; the extra respiration in the muscles releases heat. Sweating cools the body, dilation of arterioles to the skin increases heat loss, and hairs lying flat trap less air, so less insulation; all three would lower body temperature further.

11 · Part 2 Supplement Homeostasis [1]

Answer: B

Worked solution

Many students wrongly say that the capillaries move down. They do not move at all. What changes is the width of the arterioles that feed them: narrow arterioles mean little blood reaches the surface capillaries, so less heat is carried to the skin surface to be lost.

12 · Part 2 Core Tropic responses [1]

Answer: C

Explanation

The seedlings are in the dark, so the stimulus is gravity, and the roots grow downwards, towards gravity. Light cannot be the stimulus in a dark cupboard, and growing downwards is towards gravity, not away from it.

13 · Part 2 Core Tropic responses [1]

Answer: B

Explanation

Phototropism is growth towards or away from the direction of a light source; shoots bending towards a window are growing towards the light. Roots growing down and shoots growing up in the dark are responses to gravity (gravitropism), and coiling around a stick is a response to touch, not light.

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14 · Part 2 Core Tropic responses [1]

Answer: A

Explanation

In the dark the only directional stimulus is gravity. Roots grow towards gravity and shoots grow away from it, so every root curves to grow downwards and every shoot curves to grow upwards whatever position the seed started in. Roots and shoots do not just carry on in their starting direction, roots do not grow upwards, and in a dark box there is no light to make shoots grow towards one side.

15 · Part 2 Supplement Tropic responses [1]

Answer: D

Worked solution

Step 1: light from the left means the right side is shaded, so more auxin is found on the right. Step 2: auxin makes cells elongate, so the right side grows longer. Step 3: when one side grows longer than the other, the shoot curves away from the longer side, so it bends to the left, towards the light.

16 · Part 2 Core Hormones [4]

Mark scheme

(a) **[3]**

- chemical substance (1)
- produced by a gland (1)
- carried by the blood and alters the activity of one or more specific target organs (1)

ALLOW target cells / target tissues for target organs

REJECT enzyme as the description of a hormone

(b) **[1]**

- amylase is not carried by the blood / it travels in a duct instead of in the blood (1)

IGNORE amylase is an enzyme, on its own

Worked solution

Any hormone is a chemical substance made by a gland. It is carried by the blood and changes the activity of one or more specific target organs. Insulin fits all of these: it is made by the pancreas, travels in the blood and acts on the liver. Amylase is also made by the pancreas, but it travels along a duct straight into the small intestine and never enters the blood, so it does not fit the description of a hormone.

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17 · Part 2 Core Hormones [6]

Mark scheme

(a) **[2]**

- Y (1)
- any one reason from the data: Y takes longer to start (3 s compared with 0.2 s) / Y lasts longer (300 s compared with 0.5 s) (1)

ALLOW reason given without figures if the comparison is clear

- the reason mark can only be given if Y is identified

(b) **[2]**

- hormone has to be secreted into / carried by the blood to the target organ (1)
- nervous response uses (electrical) impulses that travel (quickly) along neurones (1)

ALLOW blood travels more slowly than impulses along neurones, for either mark

(c) **[2]**

- X: less damage to the hand / skin is burned less / less time in contact with the hot tray (1)
- Y: heart rate stays high while the danger may still be there / so the person can keep running or fighting / more oxygen or glucose delivered to muscles for longer (1)

Worked solution

Nervous responses start very quickly and are short-lived, while hormonal responses start more slowly and last longer. Response Y starts after 3 s and lasts 300 s, while X starts after 0.2 s and lasts only 0.5 s, so Y is the hormonal one. Hormones are slower to act because they must be secreted into the blood and carried around the body to the target organ, whereas nerve impulses travel rapidly along neurones. A fast reflex pulls the hand away before the skin is badly burned. A long-lasting rise in heart rate keeps oxygen and glucose flowing to the muscles while the person may still be in danger.

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18 · Part 2 Supplement Hormones [6]

Mark scheme

(a) **[2]**

- increase = 112 − 70 = 42 (beats per minute) (1)
- 42 ÷ 70 × 100 = 60 (%) (1)

ALLOW correct final answer with no working for 2 marks

ECF correct percentage from an incorrect increase

(b) **[3]**

- more glucose in the blood (1)
- faster heart rate means blood flows faster / more blood is pumped to the muscles (1)
- more glucose (and oxygen) delivered to muscle cells (1)
- more respiration in the muscle cells (1)
- more energy released for muscle contraction / for running (1)
- max 3

REJECT energy is produced / made by respiration

(c) **[1]**

- glucagon (1)

Worked solution

(a) Percentage increase = (new value − original value) ÷ original value × 100. The heart rate went from 70 to 112 beats per minute, an increase of 42. Then 42 ÷ 70 × 100 = 60%. Check by working backwards: 70 × 1.6 = 112.

(b) Adrenaline raised her blood glucose concentration from 88 to 108 mg per 100 cm^3, so more glucose is available. Her faster heart rate pumps blood more quickly to her muscles, delivering glucose and oxygen faster. Muscle cells use the glucose and oxygen in respiration, releasing more energy for muscle contraction so she can run.

(c) The pancreas secretes glucagon, which also raises blood glucose concentration (by making the liver convert glycogen to glucose).

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19 · Part 2 Core Hormones [5]

Mark scheme

(a) **[1]**

- in the blood / in the bloodstream / in the plasma (1)

(b) **[1]**

- more light enters the eye, so the cat can see (the dog) better in dim light (1)

(c) **[2]**

- more oxygen taken into the lungs / absorbed into the blood (1)
- more (aerobic) respiration in the muscles / more energy released for muscle contraction (1)
- more carbon dioxide removed (from the blood / muscles) (1)
- max 2

(d) **[1]**

- adrenaline remains in the blood for some time / hormonal effects last longer than nervous effects (1)

Worked solution

Adrenaline is a hormone, so it is carried in the blood. Wider pupils let more light into the eye, which helps the cat see at dusk. Faster breathing brings more oxygen into the blood, so the muscles can carry out more aerobic respiration and release more energy for running. The heart rate stays high after the danger has gone because hormonal effects last longer than nervous effects: the adrenaline is still in the blood.

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20 · Part 2 Core Homeostasis [4]

Mark scheme

(a) **[1]**

- maintenance of a constant internal environment (1)

> **ALLOW** keeping conditions inside the body constant

> **REJECT** keeping the external environment constant

(b) **[1]**

- 55 mg per 100 cm^3 (1)
- the unit is needed for the mark

(c) **[1]**

- glucose from the digested meal / starch digested to glucose / glucose absorbed into the blood (from the small intestine) (1)

(d) **[1]**

- insulin (1)

Worked solution

Homeostasis is the maintenance of a constant internal environment. The increase is 140 − 85 = 55 mg per 100 cm^3 (check: 85 + 55 = 140). The rice contains starch, which is digested to glucose and absorbed into the blood, so blood glucose rises. The pancreas then secretes insulin, which decreases the blood glucose concentration back towards normal.

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21 · Part 2 Supplement Homeostasis [5]

Mark scheme

(a) **[1]**

- 37.0 °C (1)

ALLOW 37 °C

(b) **[3]**

- rise in temperature (above the set point) is detected by the brain (1)
- brain causes responses that cool the body, e.g. sweating / vasodilation (1)
- temperature falls back to the set point (1)
- when the set point is reached, the cooling responses decrease / stop (1)
- max 3

(c) **[1]**

- heat was being released (by respiration in the muscles) faster than it could be lost (1)

ALLOW heat gain greater than heat loss

Worked solution

Before and after the exercise her temperature was steady at 37.0 °C, so this is the set point. During exercise her muscles respired faster and released heat faster than her body could lose it, so her temperature rose even though cooling responses had started. In negative feedback, the brain detects the rise above the set point and triggers responses that reverse it, such as sweating and vasodilation. Her temperature then falls back to 37.0 °C, and once it reaches the set point the cooling responses are reduced or switched off, so the temperature does not fall too low.

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22 · Part 2 Supplement Homeostasis [6]

Mark scheme

(a) **[2]**

- insulin concentration decreased (from 14 to 5 arbitrary units) (1)
- glucagon concentration increased (from 20 to 66 arbitrary units) (1)

ALLOW both levelled off / changed very little from 8 hours

(b) **[3]**

- glucagon is carried in the blood to the liver (1)
- liver converts glycogen to glucose (1)
- glucose is released into the blood (1)
- blood glucose concentration is kept at / near the set point (1)
- max 3

REJECT glucagon converted to glucose / glucagon converted to glycogen

(c) **[1]**

- less glucose is removed from the blood / less glucose is converted to glycogen (so blood glucose does not fall further) (1)

ALLOW so insulin does not cancel out the effect of glucagon

Worked solution

As time without food increased, the insulin concentration fell from 14 to 5 arbitrary units, while the glucagon concentration rose from 20 to 66 arbitrary units. Glucagon travels in the blood to the liver, where it causes stored glycogen to be converted into glucose. This glucose is released into the blood, so the blood glucose concentration levels off at about 83 mg per 100 cm^3 instead of falling further. Insulin has the opposite effect: it causes the liver to take glucose out of the blood and store it as glycogen. Keeping insulin low while food is not available means glucose is not removed from the blood when it is needed by the body's cells.

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23 · Part 2 Supplement Homeostasis [5]

Mark scheme

(a) **[3]**

- insulin injections (1)
- regular measuring / monitoring of the blood glucose concentration (1)
- control of diet, e.g. limiting carbohydrate / sugar intake or eating regular meals (1)

ALLOW insulin pump for insulin injections

- adjusting insulin dose or food intake when exercising (1)
- max 3

(b) **[2]**

- insulin would be digested (in the stomach / small intestine) (1)
- by protease (enzymes) into amino acids (1)
- so it would no longer work / would not be absorbed into the blood as insulin (1)
- max 2

Worked solution

People with Type 1 diabetes cannot make enough insulin, so their blood glucose concentration can rise too high. Treatment involves injecting insulin, regularly checking the blood glucose concentration so the dose can be adjusted, and controlling the diet, especially the amount of carbohydrate eaten. Insulin is a protein, so if it were swallowed, protease enzymes in the stomach and small intestine would digest it into amino acids. It would then no longer work as insulin, which is why it is injected instead.

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24 · Part 2 Supplement Homeostasis [6]

Mark scheme

(a) **[1]**

- brain (1)

ALLOW hypothalamus

(b) **[2]**

- muscles contract (and relax) rapidly / repeatedly (1)
- respiration (in the muscles) increases, releasing heat (1)

REJECT heat is made / produced from nothing, or muscles make energy

(c) **[2]**

- (thicker) layer of air is trapped between the hairs (1)
- air is a poor conductor of heat / is an insulator (1)

(d) **[1]**

- water in sweat evaporates (from the skin), which removes heat from the skin (1)

ALLOW evaporation of sweat cools the skin

Worked solution

The brain detects the fall in blood temperature and coordinates the responses. Shivering is rapid, repeated contraction of muscles; this increases respiration in the muscle cells, which releases heat and warms the body. When hair erector muscles contract, the hairs stand up and trap a thicker layer of air next to the skin. Air is a poor conductor of heat, so this layer insulates the body and reduces heat loss. Sweating cools the skin because the water in sweat evaporates, taking heat away from the skin. On a cold, windy hilltop this still happens, so the damp skin loses heat quickly.

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25 · Part 2 Supplement Homeostasis [6]

Mark scheme

(a) **[2]**

- as air temperature increases, blood flow through the skin increases (1)
- correct use of data, e.g. from 4 arbitrary units at 5 °C to 52 arbitrary units at 35 °C / increase is smallest at low temperatures and greatest between 20 °C and 30 °C (1)

ALLOW positive correlation

(b) **[3]**

- vasoconstriction (1)
- arterioles supplying the skin (surface capillaries) become narrower (1)
- less blood flows through the skin surface capillaries (1)
- less heat lost (from the blood) by radiation (1)
- max 3

REJECT capillaries move / capillaries constrict, as the only mechanism

(c) **[1]**

- vasodilation, so more blood flows through skin surface capillaries and more heat is lost (by radiation) (1)

Worked solution

The table shows that blood flow through the fingertip skin rises as the air gets warmer, from 4 arbitrary units at 5 °C to 52 arbitrary units at 35 °C. In the cold, vasoconstriction happens: the arterioles supplying the skin surface capillaries narrow, so less warm blood flows close to the skin surface and less heat is lost to the air by radiation. The capillaries themselves do not move. In the heat, vasodilation widens these arterioles, so more blood flows through the skin surface capillaries and more heat is lost, helping to bring body temperature back down to the set point.

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26 · Part 2 Core Tropic responses [4]

Mark scheme

(a) **[1]**

- a response in which parts of a plant grow towards or away from gravity (1)

(b) **[1]**

- a response in which parts of a plant grow towards or away from the direction of the light source (1)

> **REJECT** answers that only describe plants needing light, with no growth direction

(c) **[2]**

- root: to reach water / mineral ions / for anchorage in the soil (1)
- shoot: to reach light (above the soil) for photosynthesis (1)

Worked solution

Gravitropism is a growth response towards or away from gravity, and phototropism is a growth response towards or away from the direction of the light source. A root that grows downwards into the soil can absorb water and mineral ions and anchors the plant. A shoot that grows upwards reaches the surface and the light, so its leaves can photosynthesise.

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27 · Part 2 Core Tropic responses [5]

Mark scheme

(a) **[1]**

- phototropism (1)

(b) **[1]**

- light came from directly above / light was not from one side only (1)

ALLOW light reached all sides of the shoots equally

(c) **[1]**

- gravitropism (1)

(d) **[2]**

- any two from: type / species of seedling; temperature; volume of water given / moisture of compost; time / number of days; number of seedlings; type of compost or growing medium; size of box (2)

ALLOW age of seedlings

- 1 mark for each correct variable

Worked solution

In box P, light came from one side only and the shoots bent towards it: this is phototropism. In box Q, light came from directly above, so it was not coming from one side and there was nothing to make the shoots bend. In box R there was no light, so the shoots grew upwards, away from gravity: this is gravitropism. To make it a fair comparison, only the light conditions should differ between the boxes, so things like the type of seedling, the temperature, the amount of water and the time allowed must be kept the same.

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28 · Part 2 Core Tropic responses [5]

Mark scheme

(a) **[2]**

- root curved / grew downwards (1)
- shoot curved / grew upwards (1)

(b) **[2]**

- the clinostat turns, so gravity acts on all sides of the root and shoot equally / no side faces downwards for long (1)
- so there is no one-sided stimulus and no gravitropism / no bending (1)

(c) **[1]**

- to prevent light from one direction causing a (phototropic) response / so that the only stimulus being tested is gravity (1)

Worked solution

Seedling Y was still, so gravity acted from one direction. Its root grew towards gravity and curved downwards, and its shoot grew away from gravity and curved upwards. The clinostat with seedling X kept turning, so gravity acted on every side of the root and shoot in turn. There was no one-sided stimulus, so neither the root nor the shoot bent and both kept growing horizontally. Both were kept in the dark so that light from one side could not cause phototropism and affect the results.

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29 · Part 2 Supplement Tropic responses [6]

Mark scheme

(a) **[4]**

- auxin is made in the shoot tip (1)
- auxin diffuses (down) through the shoot from the tip (1)
- auxin is unequally distributed / more auxin on the shaded side (1)
- auxin stimulates cell elongation (1)
- cells on the shaded side elongate more, so the shoot bends towards the light (1)
- max 4

(b) **[1]**

- no tip so no auxin made / no auxin to stimulate cell elongation (1)

(c) **[1]**

- light could not reach the tip, so auxin was distributed equally (on both sides) / no more auxin on one side (1)

ALLOW tip could not detect the direction of light

Worked solution

Auxin is made in the shoot tip and diffuses down through the shoot. When light comes from one side, auxin becomes unequally distributed, with more on the shaded side. Auxin stimulates cell elongation, so cells on the shaded side elongate more than cells on the lit side, and the shoot bends towards the light (group 1). When the tip is cut off, no auxin is made, so the cells do not elongate and the shoot neither grows longer nor bends (group 2). When the tip is covered, light from one side cannot reach it, so the auxin is spread evenly; all sides elongate equally and the shoot grows straight up (group 3).

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30 · Part 2 Supplement Tropic responses [6]

Mark scheme

(a) **[2]**

- difference = 58 − 40 = 18 (μm) (1)
- 18 ÷ 40 × 100 = 45 (%) (1)

ALLOW correct final answer with no working for 2 marks

ECF correct percentage from an incorrect difference

(b) **[3]**

- auxin is unequally distributed in response to gravity / more auxin on the lower side (1)
- auxin stimulates cell elongation (1)
- cells on the lower side elongate more (than on the upper side) (1)
- lower side grows longer, so the shoot curves upwards / away from gravity (1)
- max 3

(c) **[1]**

- the response is controlled by a chemical / auxin / a plant hormone (1)

ALLOW plant growth substance / plant growth regulator

Worked solution

(a) Percentage difference = (larger − smaller) ÷ smaller × 100. Here 58 − 40 = 18 μm, and 18 ÷ 40 × 100 = 45%. Check by working backwards: 40 × 1.45 = 58.

(b) When a shoot lies on its side, auxin becomes unequally distributed in response to gravity, with more on the lower side (65 compared with 35 arbitrary units). Auxin stimulates cell elongation, so the cells on the lower side become longer (58 μm compared with 40 μm). Because the lower side grows longer than the upper side, the shoot curves upwards, away from gravity.

(c) The growth is controlled by a chemical, auxin, rather than by nerves, so it is chemical control of plant growth.

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